

STATISTICS 101 · RECITATION HOMEWORK

Homework 1: What's Typical?

Name: _____

Date: _____

Instructions: About 20 minutes. Mean = add them all, divide by how many. Range = biggest - smallest. Write your adding and dividing – the steps count. Total: **21 points**.

Part A — Mean and range

1. Five babies in the unit have heart rates of 140, 150, 130, 160, and 145 beats per minute.

(a) What is the total? (b) What is the mean? (c) What is the range? (3 pts)

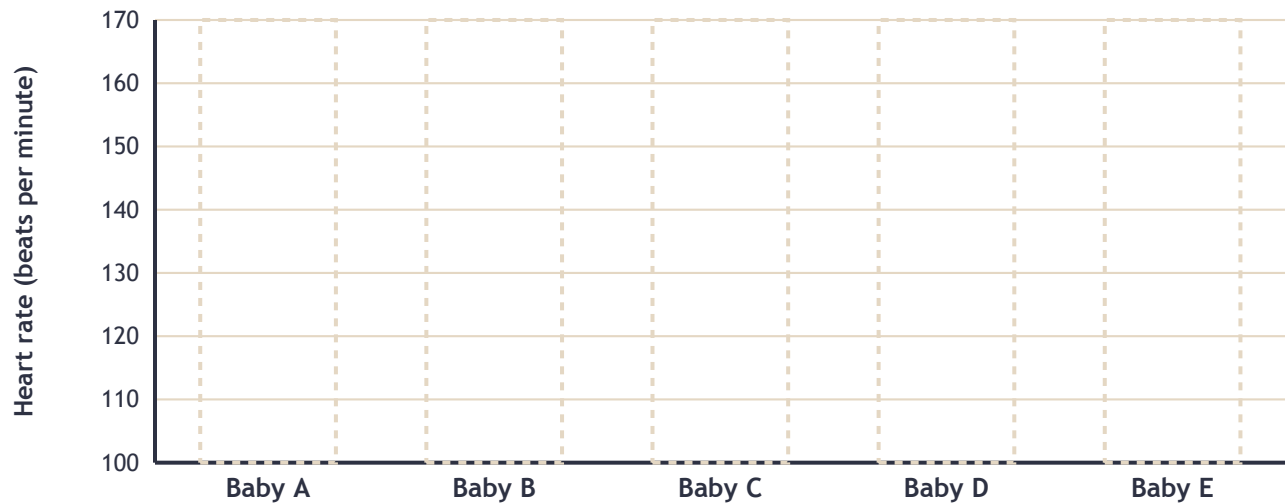
2. Four babies weigh 900 g, 1,100 g, 1,000 g, and 1,000 g. What is the mean weight? What is the range? (2 pts)

Part B — Score three babies with Apgar

3. Each baby's five Apgar signs have already been scored (0, 1, or 2 each). Add up the total, then write the category: 7-10 = doing well, 4-6 = may need some help, 0-3 = needs help right now. (6 pts)

Baby	Heart rate	Breathing	Muscle tone	Reflex	Color	Total	Category
Rosa	2	2	1	2	2		
Kai	1	1	1	1	0		
Amara	2	1	2	1	2		

Part C — Draw the data



4. On the chart above, draw a bar for each baby's heart rate from Part A: A = 140, B = 150, C = 130, D = 160, E = 145. Then answer: which baby's bar is tallest, and which bar is closest to the mean? (3 pts)

5. Two units both report an average Apgar of 7. What does the average alone NOT tell you? (1 pt)

☐
☐
☐
☐

- A. Whether the scores were all close to 7 or a mix of very high and very low.
- B. Whether the scores were added up correctly before dividing.
- C. Whether the babies were scored by a doctor or by a nurse.
- D. Whether the unit is in a big hospital or a small one.

Part D — Flashback

6. From NICU Tutor: what is the normal heart rate range for a newborn, in beats per minute? (1 pt)

7. From NICU Tutor: what does the word bradycardia mean? (1 pt)

8. From Calculus: rate of change = _____ ÷ _____. (1 pt)

Part E — Think like a doctor

9. A new nurse says, “This baby's Apgar was 5 at one minute, so he'll probably have problems growing up.” What would Dr. Virginia Apgar herself have said? (Think about what the score is FOR.) (3 pts)

Score: _____ / 21 points

Professor's comment:
